DeenGuide: An AI-Powered Quran & Sunnah Knowledge Assistant

Helps users explore Islamic knowledge through verified Quranic verses, authentic Hadith and approved reference documents, with citations and human-reviewed knowledge updates.

Name	Samiul Hasan
Student ID	C223161
GitHub username	samiul-hasan21
Email	samiulhasan226@gamil.com
Submission date	10 November 2026

AI/LLM Application Builder · CSE Industrial Training 2026 · Grow with Data

1. Problem and users

People who want to understand an Islamic topic often have to search separate Quran translations, Hadith collections and explanatory documents. They may not know the right Arabic or English keywords, exact source reference, or whether a quotation circulating online is authentic. A general-purpose chatbot can make this harder if it produces a confident answer without a verifiable source, mixes translations or editions, or invents a verse or Hadith reference.

The main users are general learners, students and people looking for introductory information about Quranic themes and the Sunnah. DeenGuide is an educational retrieval assistant. It is not intended to replace qualified scholars, issue independent fatwas, or provide authoritative rulings. Complex or disputed questions will be referred to a qualified scholar.

Why an LLM? Keyword search can find exact words but may miss a question expressed in everyday language. A static FAQ cannot naturally handle follow-up questions. An LLM can interpret a question, select retrieval tools, combine relevant passages and explain them in plain language. However, source text must come from a curated knowledge base: the model is not the authority for Quran or Hadith wording. RAG supplies evidence and citations; tools perform retrieval; guardrails limit unsupported claims.

2. Proposed solution

DeenGuide is a conversational knowledge assistant that answers questions using a local, curated collection of Quranic verses, selected Sahih al-Bukhari and Sahih Muslim Hadith, and approved reference documents. It displays source details alongside its response and states when the available collection does not support an answer.

Core user stories:

- As a learner, I want to ask a question in ordinary English and receive relevant Quranic verses with Surah and ayah references, so that I can inspect the original source.
- As a learner, I want to find relevant Hadith from selected authentic collections with collection and reference details, so that I can verify the source.
- As a user, I want to ask a follow-up such as “What does the first verse mean?” and have the assistant understand the previous answer, so that I do not need to repeat the reference.
- As a knowledge-base administrator, I want to upload a proposed reference document and review its source metadata before it becomes searchable, so that unreviewed material is not added to the shared collection.
- As a user, I want the assistant to say when it cannot find reliable supporting material, rather than guessing or presenting an unsupported religious ruling.

Example conversation (illustrative; exact wording and references will be retrieved from selected source files):

User: What guidance can I find about patience during hardship?

Assistant: (tools: search_quran, search_hadith, search_reference_docs) I found relevant passages in the indexed collection. Here are the retrieved Arabic text, selected translation, Surah and ayah reference, and matching Hadith with collection metadata. The explanation is limited to the retrieved sources.

User: What does the first verse mean?

Assistant: (rewrite_query uses conversation history) You are asking about the first Quranic reference in my previous answer. I will retrieve its approved explanatory material and show the document citation.

User: Is this a definite ruling for my personal situation?

Assistant: I can provide sources available in this collection, but I cannot issue a personal fatwa. Please consult a qualified scholar who can consider the full circumstances.

User: Add this newly uploaded Tafsir PDF to the shared knowledge base.

Assistant: (the graph pauses at interrupt()) The document has been extracted and proposed metadata is ready for review. It will not be indexed until an authorized reviewer approves it.

Why RAG and tools, not prompting alone. The knowledge collection contains exact source text and metadata that should not be recreated from model memory. RAG returns passages from the project's own documents; tools search Quran, Hadith and reference collections separately; citation validation checks that displayed references correspond to retrieved records. Prompting alone cannot guarantee that a quotation or reference exists in the source.

3. Concept coverage matrix

The project covers all seven required core concepts. E1 and E2 are the two committed electives; E3 is a stretch feature if the core application is complete.

ID	Concept	Where it appears in my project	How I'll show it at the demo
C1	Prompts and structured output	Separate prompts for classification, query rewriting, retrieval evaluation, answer generation and safety. Pydantic response includes answer, source records, confidence, evidence status and escalation flag.	Show JSON beside the chat answer; inspect prompt and schema.
C2	LangChain model layer and tracing	models.py configures Gemini as primary and an OpenAI chat model as fallback through LangChain. LangSmith traces model calls, tools, retrieval and graph execution.	Demonstrate a controlled primary-provider failure and show fallback execution and LangSmith trace.
C3	Tools and agent loop	Agent/tool layer uses search_quran, search_hadith, search_reference_docs and propose_document_ingestion. Read tools return source text and metadata; indexing is restricted.	Ask a question and show selected search tools and returned source records in the trace.
C4	Conversation memory	LangGraph SQLite checkpointer keyed by thread_id stores conversation state and prior references for follow-up questions.	Ask "What does the first verse mean?"; reopen with the same thread_id and show retained context.
C5	Guardrails and human approval	Input scope and injection checks; retrieved text treated as data; output validation requires source-backed quotations and citations; ruling requests are escalated. Document publication pauses for reviewer approval.	Try an instruction to invent a Hadith reference; show refusal. Upload a document and demonstrate approval gate.
C6	RAG with citations	ChromaDB index built from the project's own curated Quran, Hadith and approved reference files. Chunks retain collection, reference, document, page and edition metadata.	Show retrieved chunks and citations; ask outside-corpus question and show a not-found response.
C7	LangGraph workflow	StateGraph includes input_guard, rewrite_query, classify_question, retrieve, evaluate_evidence, generate_answer, validate_answer and conditional escalation/approval paths.	Show graph diagram and a traced run, including a low-evidence branch.
E1	RAG-Fusion	Generate several query variants, retrieve for each, and combine ranked results using Reciprocal Rank Fusion before evidence evaluation.	Show original question, generated queries and merged retrieval ranking.
E2	Durable pause and resume with interrupt()	Document ingestion pauses at a reviewer gate with interrupt(); persistent checkpointer stores state; reviewer approval/rejection resumes using Command(resume=...).	Pause after validation, restart if feasible, then approve/reject and show indexing only after approval.
E3	Follow-up question rewriting (stretch)	rewrite_query uses conversation history to turn context-dependent follow-ups into standalone retrieval queries. Planned as a bonus if time allows.	Show original follow-up and rewritten query in LangSmith trace.

4. Architecture

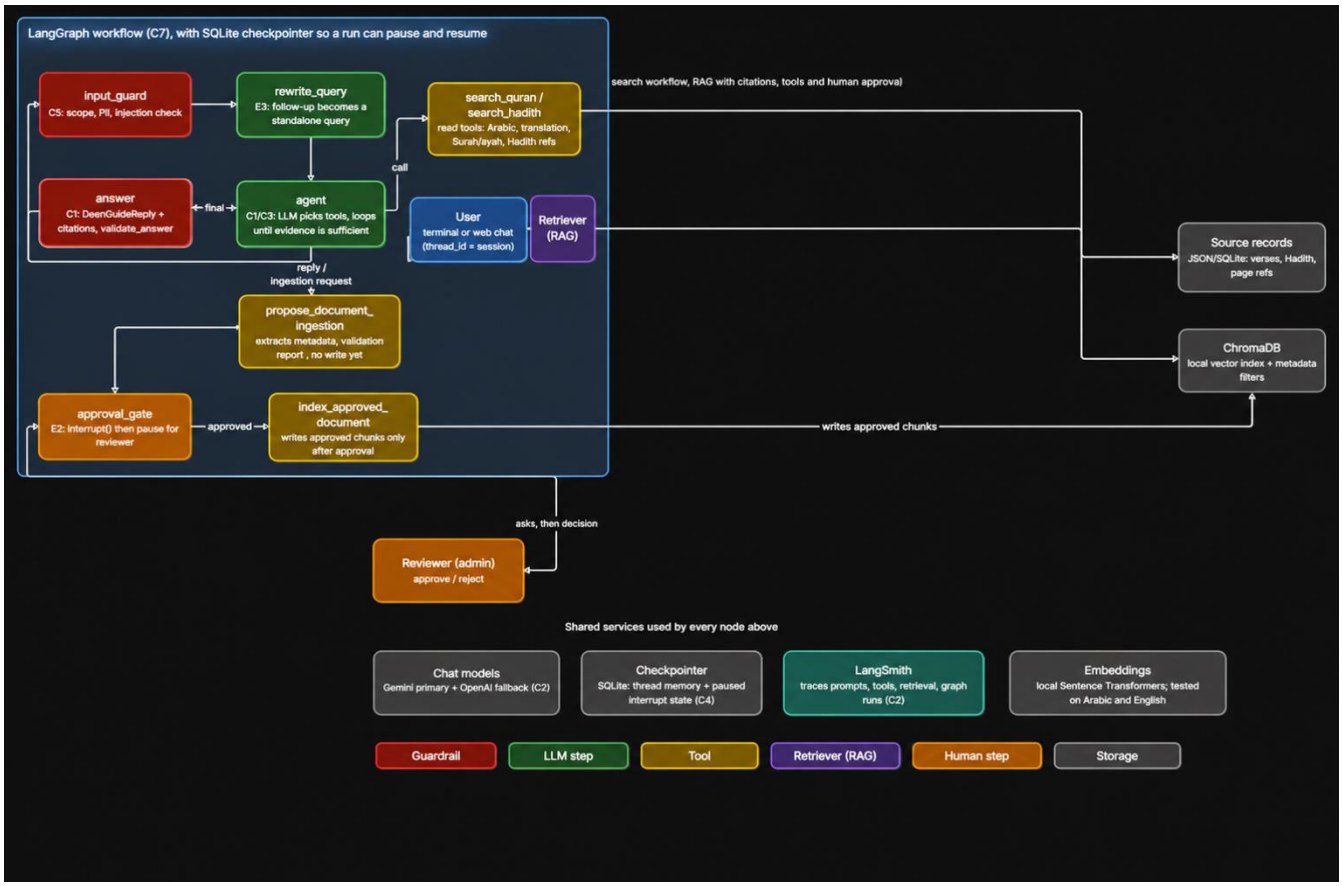


Figure 1. Proposed DeenGuide workflow. The main question-answer path is read-only; the separate knowledge-ingestion path is the only path that changes the shared searchable collection.

Node / component	Responsibility
User chat	Receives question with thread_id and conversation state.
input_guard	Checks scope, prompt injection and unsafe requests; applies refusal/escalation policy.
rewrite_query (E3)	Uses prior turns to resolve “this”, “that verse” and other follow-ups.
classify_question	Routes Quran, Hadith, general knowledge, emotional support or ruling-related questions.
RAG-Fusion retriever (E1)	Creates query variants; searches ChromaDB; merges ranked results.
evaluate_evidence	Checks relevance, source metadata and evidence sufficiency; may retrieve again or abstain.
generate_answer	Uses LangChain primary model with fallback; returns a Pydantic DeenGuideReply.
validate_answer	Checks source references, quotation provenance and unsupported claims before display.
final response	Displays answer, original source text, translation, reference and document/page citations.
Separate ingestion path	Upload → extract and inspect metadata → prepare indexing proposal → approval_gate (interrupt) → index approved document.

Shared services

- LangGraph SQLite checkpointer: stores conversation state and pending interrupt state by thread_id.
- ChromaDB: stores embeddings and metadata for the curated local knowledge corpus.
- LangChain model layer: primary Gemini model and a different-provider fallback model.
- LangSmith: traces prompts, model calls, tool calls, retrieval and workflow outcomes.

• Gradio interface: chat screen, source cards and a separate administrator review screen.

The question-answer path does not publish or modify religious source material. The ingestion path is separate and requires an authorized human decision before a new reference document is indexed.

5. Data, knowledge and tools

Knowledge corpus. The first version will use a limited, auditable collection rather than attempting to index all Islamic literature. (1) Quran text with Arabic and one clearly identified English translation, including Surah and ayah metadata. (2) A selected set or complete permitted text collection of Sahih al-Bukhari and Sahih Muslim Hadith, with book/chapter and Hadith reference metadata. (3) A small set of approved Tafsir or introductory reference documents in PDF/text form, with page numbers and source/edition details. Exact sources, editions, translations and reuse permissions will be recorded before ingestion. Material with unclear permission or licensing will be omitted.

Indexing plan. Each source record will be normalized and checked before chunking. Quran verses and Hadith records will remain intact as source units rather than being split arbitrarily. Longer reference PDFs will be split by page and section. Metadata will include source type, collection, Surah/ayah or book/Hadith number, document title, author/translator where known, edition, page, language and provenance note. The initial target is a small test corpus sufficient for a 30–50 question evaluation set; corpus size depends on verified source availability and time.

Backend storage. ChromaDB will store embeddings and metadata locally. Original source records will also be retained in structured JSON/JSONL or SQLite so that displayed Arabic text, translation and reference are taken from the source record, not reconstructed by the LLM. The LangGraph checkpointer will use SQLite for conversation and paused-workflow state.

Existing starting point. I have an earlier Gradio-based “Reflections from the Quran & Sunnah” prototype that retrieves from a local knowledge base. I will reuse suitable interface and verified-text retrieval components where practical, then add the required LangChain model layer, LangGraph workflow, RAG-Fusion, LangSmith tracing and reviewer-controlled ingestion. The existing emotion matcher will not substitute for the new document RAG pipeline.

Tool	Input	Output	Reads or writes	Needs human approval?
search_quran	Question or query variants	Verse records: Arabic, selected translation, Surah/ayah and translation metadata	Reads local Quran collection	No
search_hadith	Question or query variants	Hadith records: text, collection, book/chapter and reference metadata	Reads local Hadith collection	No
search_reference_docs	Question or query variants	Relevant chunks with document title, edition and page number	Reads ChromaDB	No
propose_document_ingestion	Uploaded file and extracted metadata	Validation report and proposed document record/chunks; no indexing yet	Prepares write	Yes, before indexing
index_approved_document	Approved proposal ID and reviewer decision	Indexes approved chunks and records approval metadata	Writes to shared index	Yes; interrupt gate

6. Safety and human oversight

The main risks are presenting an incorrect religious quotation or reference, treating a translation or explanation as the original source, giving an unsupported ruling, or allowing unreviewed material into the shared knowledge base. DeenGuide is an educational retrieval assistant, not a mufti or a substitute for qualified scholarship. Its answers will be limited to the scope and quality of the indexed collection.

Risk	Guardrail or human check
Invented verse or Hadith	Original wording is displayed only from stored source records. The model cannot create a citation record. A validator checks that every displayed reference maps to a retrieved record; otherwise the answer is withheld or marked unsupported.
Incorrect or mismatched translation	Every translation is labelled with its translator/source and edition where available. Arabic text and translation are stored as separate fields; the system does not silently merge editions.
Unsupported religious ruling	Classify ruling-seeking questions. Provide relevant source material with a limitation notice and refer the user to a qualified scholar; do not issue a personal fatwa.
Prompt injection in user input or documents	input_guard checks requests; retrieved documents are treated as reference data, not instructions. Tool permissions and validation are enforced in code, not only prompts.
Unreviewed or malicious document ingestion	Uploaded files are isolated, parsed and previewed. The shared index changes only after an authorized reviewer approves the proposed source and metadata through interrupt().
Privacy and sensitive conversation text	Avoid collecting unnecessary personal details. Do not log raw sensitive content where it is not needed; configure tracing carefully and mask user identifiers.
Low retrieval confidence or missing source	Use an evidence threshold and “not found in the available collection” response. Do not fill missing citations from model memory.
Runaway calls or provider outage	Limit tool calls, retries, input length and rate; use a different-provider fallback; keep a recorded demo run as backup.

Human reviewer. The reviewer is the project administrator or a designated knowledgeable reviewer for the training demo. Approval only controls whether a proposed reference document enters the shared index; it does not certify a religious interpretation or make the reviewer a source of religious authority. The demo will use sample documents and a simulated reviewer account.

7. Tech stack, models and cost

Layer	Choice	Why
Chat model and provider	Gemini through Google AI Studio	Primary model for classification, query generation and grounded response drafting; choose a currently available model at implementation time.
Fallback model	OpenAI API chat model	Different provider so a Gemini outage or quota failure does not stop the demo. API use may require billing; keep fallback calls limited.
Embedding model	Local Sentence Transformers model	Runs locally and avoids a separate embedding API key; test multilingual retrieval for Arabic and English.
Vector store	ChromaDB local persistent storage	Simple local vector store for a small curated corpus and metadata filtering.
Workflow and state	LangGraph + SQLite checkpointer	Graph orchestration, conversation state and durable interrupt/resume.

Other libraries	LangChain, Pydantic, LangSmith, PyMuPDF, Gradio, SQLite	Python stack; PyMuPDF extracts page-aware PDF text; Gradio provides the demo UI.
Expected usage and cost	Small development corpus; limited test calls and demo traffic	Embeddings are generated locally and indexed once. LLM and tracing costs depend on provider quotas and usage; monitor usage and do not assume all API calls are free.

8. Risks and fallbacks

Risk	Likelihood	What I'll do if it happens
Source licensing or provenance is unclear	Medium	Use only material whose reuse terms are understood; record edition and provenance; omit questionable files.
Arabic text extraction or PDF parsing is unreliable	Medium	Prefer structured text for Quran/Hadith; inspect PDF extraction; manually verify a small reference set and retain page metadata.
Retrieval misses a relevant passage	Medium	Use RAG-Fusion, metadata filters and a manually checked evaluation set; return not-found when evidence remains insufficient.
Gemini/OpenAI quota, billing or outage	Medium	Use provider fallback, rate limits and cached/recorded demo examples; test fallback before presentation.
Durable pause/resume fails during live demo	Low to medium	Test persistence early; keep an in-process approval path as backup and record a successful pause/resume run.
Scope exceeds 2–3 weeks	Medium	Prioritize seven core concepts and two committed electives; reduce corpus size and defer bonus follow-up rewriting or UI polish.
Incorrect answer despite retrieved evidence	Medium	Validate source identifiers and quotations; evaluate faithfulness manually; show retrieved passages and label explanatory text.
User treats assistant as religious authority	Medium	Display scope limitations, avoid fatwas, and direct complex or personal rulings to qualified scholars.

References

1. LangChain documentation. Retrieval, tools, model fallbacks and structured output. https://python.langchain.com/docs/
2. LangGraph documentation. Persistence, checkpointers and human-in-the-loop interrupt/resume. https://langchain-ai.github.io/langgraph/
3. LangSmith documentation. Tracing and observability for LangChain/LangGraph applications. https://docs.smith.langchain.com/
4. Chroma documentation. Local vector database, embeddings and metadata filtering. https://docs.trychroma.com/
5. Quran Foundation API documentation. Quran text and content API reference. https://api-docs.quran.com/ (used as a source discovery/reference; the project RAG corpus will be stored locally).
6. Grow with Data, CSE Industrial Training 2026. Course materials and notebooks for Sessions 4–12: prompts, LangChain, tools, memory, safety, RAG, LangGraph, human approval, long-term memory and MCP.